

Embedded Systems Programming

College of Science and Technology

School Of ICT

Computer Science Department

Lecturer : NGABO Desire





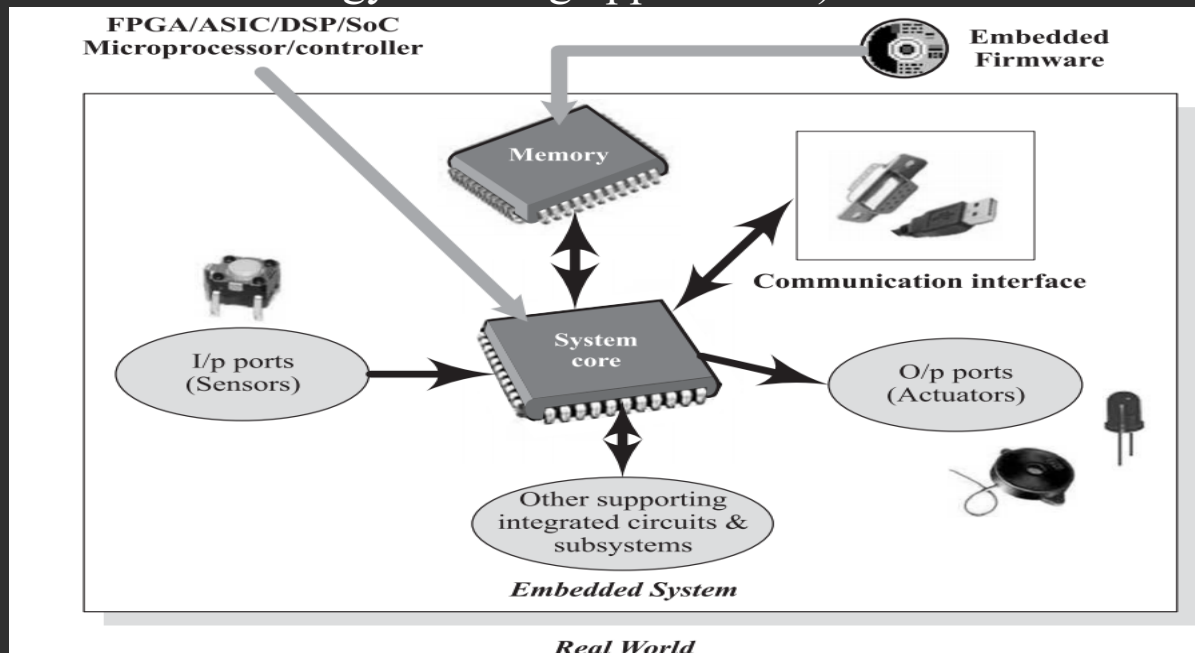
Lecture3



TYPICAL EMBEDDED SYSTEM

Overview

- ◆ A typical embedded system contains a single chip controller, which acts as the master brain of the system
- ◆ The controller can be:
 - ◆ a Microprocessor (e.g. Intel 8085) or
 - ◆ a microcontroller (e.g. Atmel AT89C51) or
 - ◆ a Field Programmable Gate Array (FPGA) device (e.g. Xilinx Spartan,A) or
 - ◆ a Digital Signal Processor (DSP) (e.g. Blackfi n® Processors from Analog Devices) or
 - ◆ an Application Specific Integrated Circuit (ASIC)/ Application Specific Standard Product (ASSP) (e.g. ADE7760 Single Phase Energy Metering IC from Analog Devices for energy metering applications)



Elements of an embedded system

CORE OF THE EMBEDDED SYTEM

- ◆ A typical embedded systems are domain and application specific and are built around a central core.
- ◆ The core of the embedded system falls into any one of the following categories:
 1. General Purpose and Domain Specific Processors
 - ◆ Microprocessors
 - ◆ Microcontrollers
 - ◆ Digital Signal Processors
 2. Application Specific Integrated Circuits (ASICs)
 3. Programmable Logic Devices (PLDs)
 4. Commercial off-the-shelf Components (COTS)
- ◆ Note that if you assess any embedded system, you will find that it is built around any of the core units mentioned above.

CORE OF THE EMBEDDED SYTEM



1. General Purpose and Domain Specific Processors

- ◆ Almost 80% of the embedded systems are processor/controller based
- ◆ A Microprocessor is a silicon chip representing a central processing unit (CPU), which is capable of performing arithmetic as well as logical operations according to a pre-defined set of instructions, which is specific to the manufacturer
- ◆ Based on the purpose they serve; processors are classified as:
 - ◆ General-Purpose Processors (GPPs)
 - ◆ and Application-Specific Instruction Set Processors (ASIPs)
- ◆ General Purpose Processor (GPP)
 - ◆ A GPP is a processor designed for general computational tasks (e.g., processors used in desktop and laptop computers). Due to mass-production their cost is low
- ◆ Application-Specific Instruction Set Processor (ASIP):
 - ◆ ASIP is a processor with architecture and instruction set optimized for a specific application

CORE OF THE EMBEDDED SYTEM



- ◆ Based on the domain and application, the processor may be a microprocessor, microcontroller or a digital signal processor (DSP)
- ◆ Most of the applications use microprocessors or microcontrollers. Some speech and video signal processing applications use DSPs
 - ◆ Microprocessors
 - ◆ Examples: Zilog Z80 (8-bit), Intel 8086 (16-bit), Motorola 68020 (32-bit)
 - ◆ Raspberry pi
 - ◆ Microcontrollers
 - ◆ Microcontrollers are more powerful and more preferred than microprocessors. They may be either general-purpose microcontrollers (Intel 8051) or domain specific microcontrollers (Atmel's AVR). TMS1000 (from Texas instruments) is world's first microcontroller. Some of Arduino board microcontroller (ATMEGA328)
 - ◆ Digital Signal Processors (DSPs)
 - ◆ They are powerful, special-purpose, 8- or 16- or 32-bit microprocessors. They are used in signal processing applications. (e.g., Speech processing, image processing, video processing, video games, digital cameras. HDTV, radar, sonar, etc.)
 - ◆ They are 2 or 3 times faster than the general-purpose microprocessors
 - ◆ They involve large number of real-time calculations
 - ◆ They implement algorithms in hardware which speeds up the execution
 - ◆ Examples for DSP: Motorola 56001

CORE OF THE EMBEDDED SYTEM



◆ Classification of Processors/Controllers:

◆ Processors/controllers are classified according to the following criteria:

1. Instruction set
2. Architecture
3. Order of storing bits

1. Based on instruction set:

◆ RISC processors/Controllers:

- ◆ RISC stands for Reduced instruction Set Computing. RISC processors/controllers have a smaller number of instructions.
- ◆ Example: ARM, AVR microprocessors
- ◆ Important applications: Security systems, home automation, etc

◆ CISC processors / Controllers:

- ◆ CISC stands for Complex Instruction Set Computing. CISC processor / controller has large and complex instruction set.
- ◆ Example: VAX, AMD microprocessors
- ◆ Important applications: Smartphones, PDAs.

CORE OF THE EMBEDDED SYTEM

❖ Classification of Processors/Controllers:

1. Based on instruction set...

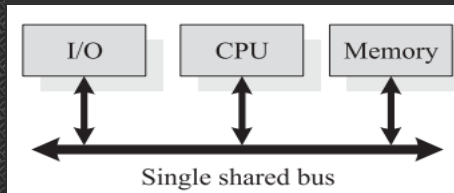
RISC	CISC
Lesser number of instructions	Greater number of Instructions
Instruction pipelining and increased execution speed	Generally no instruction pipelining feature
Orthogonal instruction set (Allows each instruction to operate on any register and use any addressing mode)	Non-orthogonal instruction set (All instructions are not allowed to operate on any register and use any addressing mode. It is instruction-specific)
Operations are performed on registers only, the only memory operations are load and store	Operations are performed on registers or memory depending on the instruction
A large number of registers are available	Limited number of general purpose registers
Programmer needs to write more code to execute a task since the instructions are simpler ones	Instructions are like macros in C language. A programmer can achieve the desired functionality with a single instruction which in turn provides the effect of using more simpler single instructions in RISC
Single, fixed length instructions	Variable length instructions
Less silicon usage and pin count	More silicon usage since more additional decoder logic is required to implement the complex instruction decoding.
With Harvard Architecture	Can be Harvard or Von-Neumann Architecture



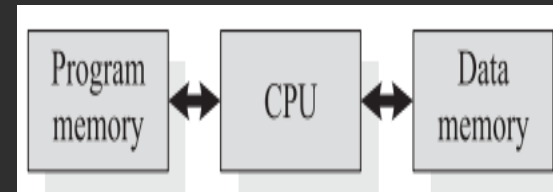
2. Architecture

- ◇ Von-Neumann architecture microprocessors/microcontrollers
 - ◇ It shares a single bus for program memory and data memory.
 - ◇ Processor first fetches instruction and then fetches data needed. Two separate fetches slow down processor operation.
 - ◇ Program memory and data memory are stored on same chip. So, there is a chance of accidental corruption of program memory.
 - ◇ Cheaper and no memory alignment problems
 - ◇ Von-Neumann architecture is also referred as Princeton architecture, since it was developed by the Princeton University.
- ◇ Harvard architecture microprocessors/microcontrollers
 - ◇ It contains separate buses for program memory and data memory. So, it has more speed.
 - ◇ Program memory and data memory are stored at different locations. So, no chances for accidental corruption of program memory
 - ◇ It is costly and has memory alignment problems

2. Architecture



Von-Neumann



Harvad

Harvard Architecture	Von-Neumann Architecture
Separate buses for instruction and data fetching	Single shared bus for instruction and data fetching
Easier to pipeline, so high performance can be achieved	Low performance compared to Harvard architecture
Comparatively high cost	Cheaper
No memory alignment problems	Allows self modifying codes [*]
Since data memory and program memory are stored physically in different locations, no chances for accidental corruption of program memory	Since data memory and program memory are stored physically in the same chip, chances for accidental corruption of program memory



2. Application Specific Integrated Circuits (ASICs)
 - ◆ ASIC is a microchip designed for a specific application
 - ◆ ASICs are pre-fabricated or customized for a special purpose
 - ◆ They are not reprogrammable
 - ◆ They consume small area and so reduce the system size.
 - ◆ On mass-production, ASICs are profitable
 - ◆ Most of the ASICs are proprietary products. So, their developers keep the internal details confidential
 - ◆ **Applications of ASICs:** Chips used for toys, handheld computers, cell phones and voice recorders
 - ◆ Example:
 - ◆ The ADE7760 Energy Meter ASIC developed by Analog Devices for Energy metering applications is a typical example for ASSP

CORE OF THE EMBEDDED SYTEM

3. Programmable Logic Devices (PLDs)

- ◆ Logic devices provide specific functions, including device-to-device interfacing, data communication, signal processing, data display, timing and control operations, and almost every other function a system must perform
- ◆ There are two types of logic devices-
 - ◆ Fixed logic devices and
 - ◆ Programmable logic devices
- ◆ Fixed logic devices contain permanent digital circuits.
 - ◆ They perform one or a set functions.
 - ◆ Once manufactured, they cannot be changed
 - ◆ Example
 - ◆ a logic gate or a multiplexer or a flip-flop performs a given logic function that is known at the time of device manufacture
- ◆ Programmable logic devices
 - ◆ Complex Programmable Logic Devices (CPLDs)
 - ◆ Example : [XC9500 CPLD](#)
 - ◆ and Field Programmable Gate Arrays(FPGAs)
 - example: [Xilinx FPGAs](#) , [Cyclone V SX SoC DE10-Standard](#)

CORE OF THE EMBEDDED SYTEM

4. Commercial off-the-shelf Components (COTS)

- ◆ A Commercial Off-the-Shelf (COTS) product is one which is used 'as-is'. COTS products are designed in such a way to provide easy integration and interoperability with existing system components
- ◆ COTS refer to hardware or software products that are ready-made and available for sale, lease, or license in the commercial market

- ◆ **Advantages of COTS:**

- ◆ Reduce development time.
- ◆ Readily available and up-to-date
- ◆ You can access reviews and advice from existing users



[W6300-EVB-Pico2](#)

- ◆ **Disadvantages of COTS:**

- ◆ Have some unwanted extra features
 - ◆ Additional licensing cost
 - ◆ Manufacturer of the product may withdraw the product or discontinue the production of COTS at any time
- ◆ Examples of COTS: MS-Office, Anti-virus software, Adobe Photoshop, etc

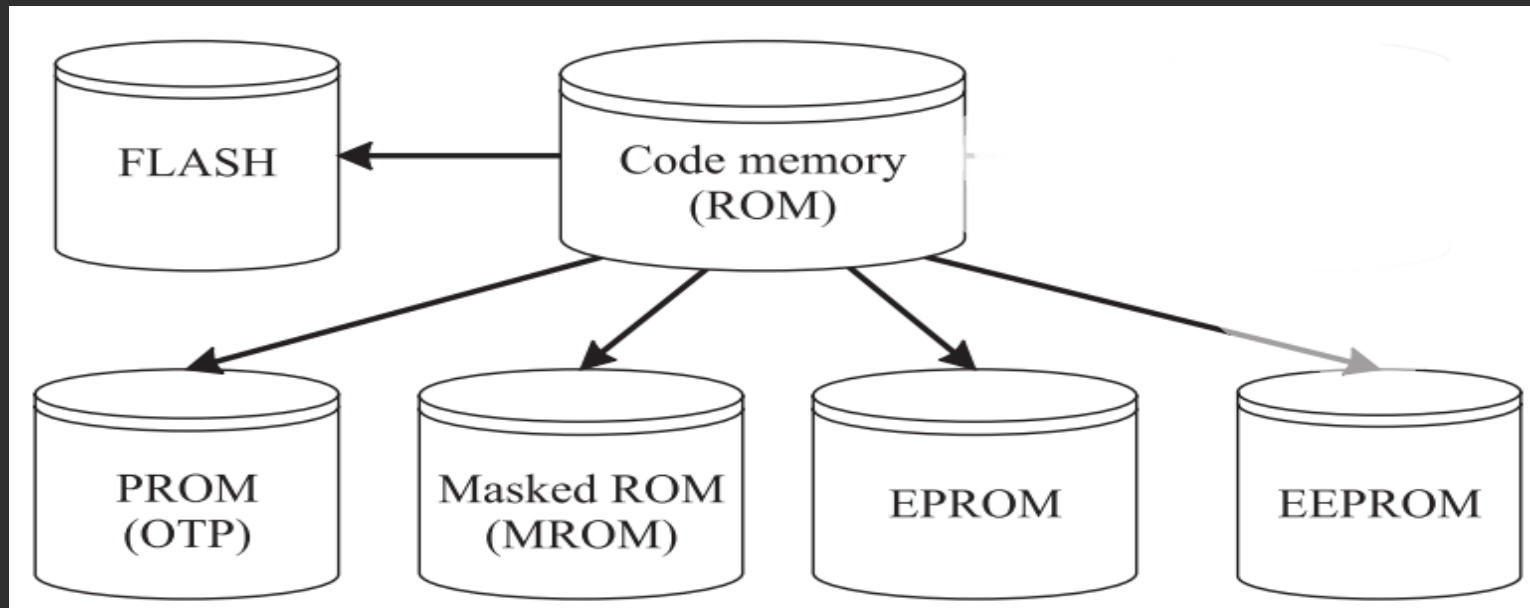
Memory



- ◆ Memory is an important part of a processor/controller based embedded systems
 - ◆ If the memory is built in processor/controller chip, it is called on-chip memory
 - ◆ If the memory is external to the processor/controller chip, it is called off-chip memory
 - ◆ On-chip memory is smaller and faster than off-chip memory
- ◆ Frequently used data is stored in on-chip memory
 - ◆ The types of on-chip memory are cache and on-chip SRAM
- ◆ Basically, memory is classified as RAM and ROM

Program Storage Memory (ROM)

- ◆ Program instructions of an embedded system are stored in the program memory or code storage memory
- ◆ The code memory retains its contents even after the power to it is turned off.
- ◆ It is generally known as non-volatile storage memory.
- ◆ Depending on the fabrication, erasing, and programming techniques they are classified into the following types as illustrated in the figure below

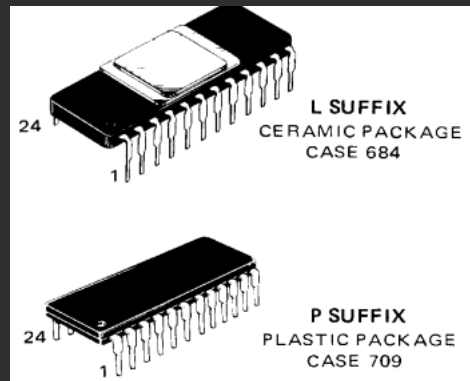


Classification of Program Memory (ROM)

Program Storage Memory (ROM)....

◆ Masked ROM (MROM)

- ◆ is a one-time programmable device.
- ◆ makes use of the hardwired technology for storing data.
 - ◆ The device is factory programmed by masking and metallisation process at the time of production itself, according to the data provided by the end user.
- ◆ The primary advantage of this is low cost for high volume production.
- ◆ They are the least expensive type of solid state memory

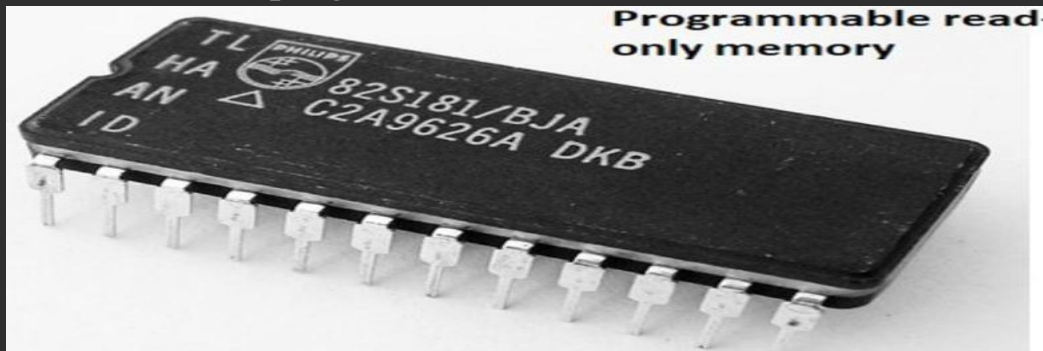


Mask Rom

Program Storage Memory (ROM)....

◆ Programmable Read Only Memory (PROM) / (OTP)

- ◆ Unlike Masked ROM Memory, One Time Programmable Memory (OTP) or PROM is not pre-programmed by the manufacturer.
- ◆ The end user is responsible for programming these devices.
- ◆ The memory has nichrome or polysilicon wires arranged in a matrix.
 - ◆ The wires can be functionally viewed as fuses
- ◆ It is programmed by a PROM programmer
- ◆ which selectively burns the fuses according to the bit pattern to be stored.
- ◆ Fuses which are not blown/burned represents a logic “1” whereas fuses which are blown/burned represents a logic “0”.
- ◆ The default state is logic “1”.
- ◆ OTP is widely used for commercial production of embedded systems whose prototyped versions are proven and the code is finalised.
- ◆ It is a low cost solution for commercial production.
- ◆ OTPs cannot be reprogrammed

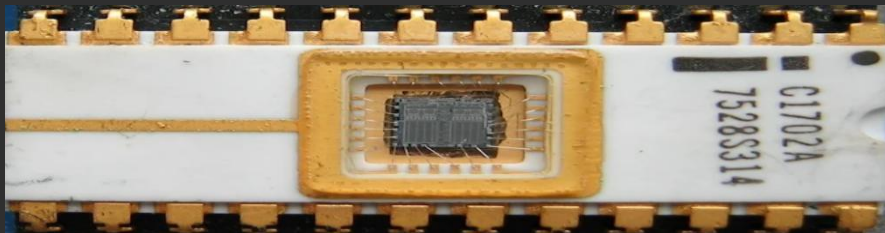


PROM

Program Storage Memory (ROM)....

◆ Erasable Programmable Read Only Memory (EPROM)

- ◆ OTPs are not useful and worth for development purpose.
 - ◆ During the development phase, the code is subject to continuous changes and using an OTP each time to load the code is not economical.
- ◆ Erasable Programmable Read Only Memory (EPROM) gives the flexibility to re-program the same chip.
- ◆ EPROM stores the bit information by charging the floating gate of a Field Effect Transistor (FET).
- ◆ Bit information is stored by using an EPROM programmer
 - ◆ It applies high voltage to charge the floating gate.
- ◆ EPROM contains a quartz crystal window for erasing the stored information.
- ◆ Note that :
 - ◆ If the window is exposed to ultraviolet rays for a fixed duration, the entire memory will be erased.
 - ◆ Even though the EPROM chip is flexible in terms of re-programmability, it needs to be taken out of the circuit board and put in a UV eraser device for 20 to 30 minutes.
 - ◆ it is a tedious and time-consuming process



Intel 2708: The EPROM

Program Storage Memory (ROM)....



- ◆ Electrically Erasable Programmable Read Only Memory (EEPROM)
 - ◆ As the name indicates, the information contained in the EEPROM memory can be altered by using electrical signals at the register/ Byte level.
 - ◆ They can be erased and reprogrammed in-circuit.
 - ◆ These chips include a chip erase mode and in this mode they can be erased in a few milliseconds.
 - ◆ It provides greater flexibility for system design.
 - ◆ The only limitation is their capacity is limited when compared with the standard ROM (A few kilobytes)



EEPROM

Program Storage Memory (ROM)....



◆ FLASH

- ◆ is the latest ROM technology and is the most popular ROM technology used in today's embedded designs
- ◆ . FLASH memory is a variation of EEPROM technology.
- ◆ It combines the reprogrammability of EEPROM and the high capacity of standard ROMs. FLASH memory is organised as sectors (blocks) or pages.
- ◆ FLASH memory stores information in an array of floating gate MOSFET transistors.
- ◆ The erasing of memory can be done at sector level or page level without affecting the other sectors or pages.
- ◆ Each sector/page should be erased before re-programming.
- ◆ The typical erasable capacity of FLASH is of the order of a few 1000 cycles.
- ◆ SST39LF010 from Microchip (www.microchip.com) is an example of 1Mbit (Organised as 128K x8) Flash memory with typical endurance of 100,000 cycles



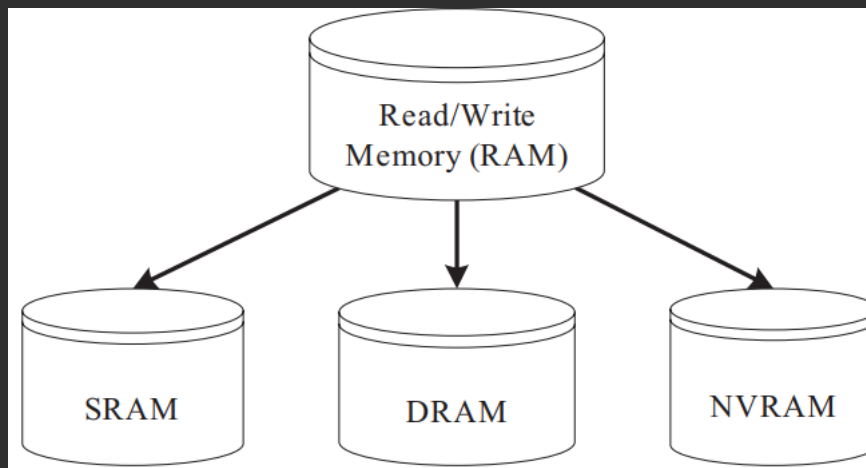
SST39LF010

Read-Write Memory/Random Access Memory (RAM)



◆ RAM

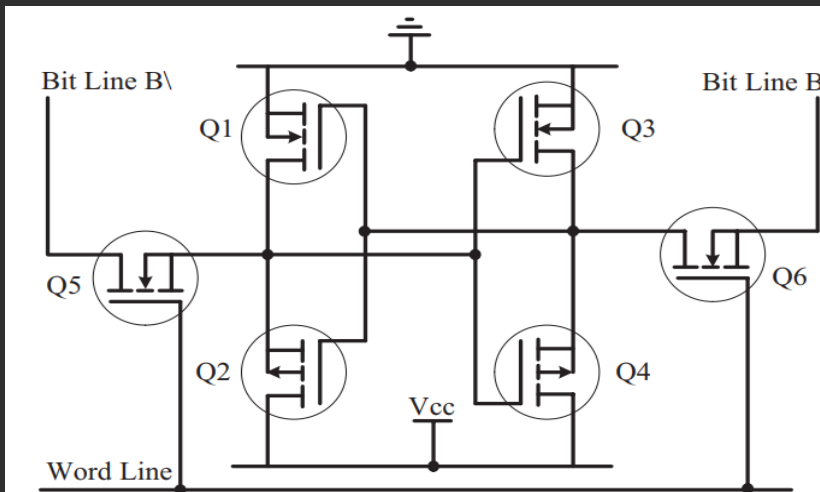
- ◆ is the data memory or working memory of the controller/processor.
- ◆ Controller/processor can read from it and write to it.
- ◆ is volatile, meaning when the power is turned off, all the contents are destroyed.
- ◆ is a direct access memory, meaning we can access the desired memory location directly without the need for traversing through the entire memory locations to reach the desired memory position (i.e. random access of memory location).
- ◆ RAM generally falls into three categories as illustrated below



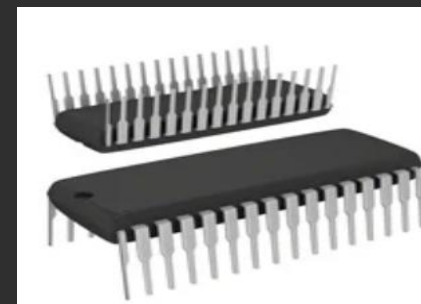
Classification of Working Memory (RAM)

Read-Write Memory/Random Access Memory (RAM)...

- ◆ Static RAM (SRAM)
 - ◆ Stores data in the form of voltage.
 - ◆ They are made up of flipflops.
 - ◆ is the fastest form of RAM available.
 - ◆ In typical implementation, an SRAM cell (bit)
 - ◆ is realised using six transistors (or 6 MOSFETs).
 - ◆ Four of the transistors are used for building the latch (flipflop) part of the memory cell and two for controlling the access.
 - ◆ SRAM is fast in operation due to its resistive networking and switching capabilities.
 - ◆ In its simplest representation an SRAM cell can be visualised



SRAM cell implementation

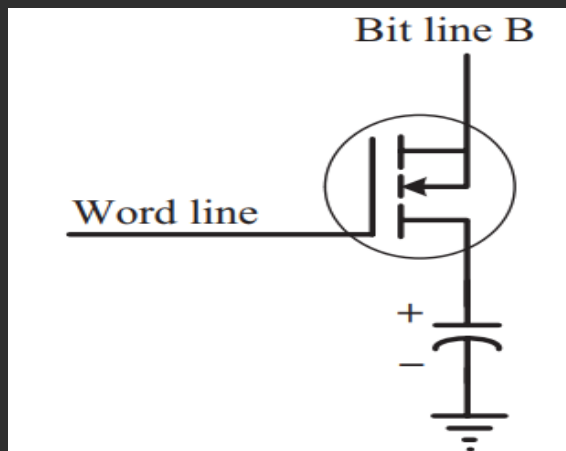


IC SRAM

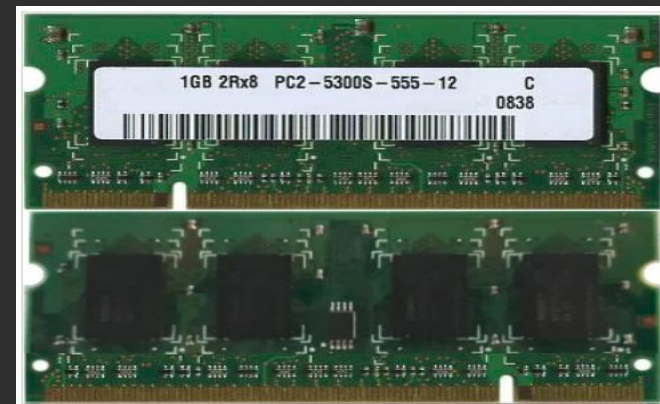
Read-Write Memory/Random Access Memory (RAM)...

◆ Dynamic RAM (DRAM)

- ◆ Dynamic RAM stores data in the form of charge.
- ◆ They are made up of MOS transistor gates.
- ◆ The advantages of DRAM are its high density and low cost compared to SRAM.
- ◆ The disadvantage is that since the information is stored as charge it gets leaked off with time and to prevent this they need to be refreshed periodically.
- ◆ Special circuits called DRAM controllers are used for the refreshing operation.
- ◆ The refresh operation is done periodically in milliseconds interval
- ◆ Referring to the cell illustration below
 - ◆ The MOSFET acts as the gate for the incoming and outgoing data whereas the capacitor acts as the bit storage unit



DRAM cell implementation



IC DRAM

Read-Write Memory/Random Access Memory (RAM)...

Static Ram VS Dynamic RAM (DRAM)



SRAM cell	DRAM cell
Made up of 6 CMOS transistors (MOSFET)	Made up of a MOSFET and a capacitor
Doesn't require refreshing	Requires refreshing
Low capacity (Less dense)	High capacity (Highly dense)
More expensive	Less expensive
Fast in operation. Typical access time is 10ns	Slow in operation due to refresh requirements. Typical access time is 60ns. Write operation is faster than read operation.

Read-Write Memory/Random Access Memory (RAM)...



- ◆ Static NVRAM (Non-volatile RAM)
 - ◆ is a random access memory with battery backup.
 - ◆ It contains static RAM based memory and a minute battery for providing supply to the memory in the absence of external power supply.
 - ◆ The memory and battery are packed together in a single package.
 - ◆ NVRAM is used for the nonvolatile storage of results of operations or for setting up of flags, etc.
 - ◆ The life span of NVRAM is expected to be around 10 years.
 - ◆ DS1744 from Maxim/Dallas is an example for 32KB NVRAM

Memory According to the Type of Interface

- ◆ The interface (connection) of memory with the processor/controller can be of various type
 - ◆ It may be a parallel interface [The parallel data lines (D0-D7) for an 8 bit processor/controller will be connected to D0-D7 of the memory] or
 - ◆ The interface may be a serial interface like I2C (Pronounced as I Square C)
 - ◆ or it may be an SPI (Serial peripheral interface, 2+n line interface where n stands for the total number of SPI bus devices in the system)
 - ◆ Serial interface is commonly used for data storage memory like EEPROM
 - ◆ The memory density of a serial memory is usually expressed in terms of kilobits,
 - ◆ whereas that of a parallel interface memory is expressed in terms of kilobytes.
 - ◆ Atmel Corporations AT24C512 is an example for serial memory with capacity 512 kilobits and 2-wire interface.
 - ◆ For communication interface (I2C, SPI, ..) you can refer to next lecture slides

Memory Shadowing

- ❖ The execution of a program or a configuration from a Read Only Memory (ROM) is very slow (120 to 200 ns) compared to the execution from a random access memory (40 to 70 ns).
 - ❖ From the timing parameters it is obvious that RAM access is about three times as fast as ROM access.
- ❖ Shadowing of memory is a technique adopted to solve the execution speed problem in processor-based systems
- ❖ In computer systems and video systems there will be a configuration holding ROM called Basic Input Output Configuration ROM or simply BIOS
- ❖ In personal computer systems BIOS stores the hardware configuration information like the address assigned for various serial ports and other non-plug 'n' play devices
 - ❖ it is read and the system is configured according to it during system boot up and it is time consuming
- ❖ Currently manufactures included a RAM behind the logical layer of BIOS at its same address as a shadow to the BIOS and the first step that happens during the boot up is copying the BIOS to the shadowed RAM and write protecting the RAM then disabling the BIOS reading

Memory Selection for Embedded Systems



- ◆ Embedded systems require a program memory
- ◆ for holding the control algorithm (For a super-loop based design) or
- ◆ embedded OS and the applications designed to run on top of it (for OS based designs),
- ◆ data memory for holding variables and temporary data during task execution, and memory for holding non-volatile data (like configuration data, look up table etc) which are modifiable by the application (Unlike program memory, which is non-volatile as well unalterable by the end user)
- ◆ The memory requirement for an embedded system in terms of RAM and ROM (EEPROM/FLASH/NVRAM) is solely dependent on the type of the embedded system and the applications for which it is designed
- ◆ There is no hard and fast rule for calculating the memory requirements
 - ◆ Example 1 :
 - ◆ if the embedded system is designed using SoC or a microcontroller with on-chip RAM and ROM (FLASH/EEPROM), depending on the application need the on-chip memory may be sufficient for designing the total system.
- ◆ As a rule of thumb, identify your system requirement and based on the type of processor (SoC or microcontroller with on-chip memory) used for the design, take a decision on whether the on-chip memory is sufficient or external memory is required

Memory Selection for Embedded Systems



- ◆ Example2
 - ◆ a simple electronic toy design
 - ◆ As the complexity of requirements are less and data memory requirement are minimal, we can think of a microcontroller with a few bytes of internal RAM, a few bytes or kilobytes (depending on the number of tasks and the complexity of tasks) of FLASH memory and a few bytes of EEPROM (if required) for designing the system
 - ◆ There is no need for external memory at all
- ◆ If the embedded design is based on an RTOS(Real time operating System),
 - ◆ RTOS requires certain amount of RAM for its execution and
 - ◆ ROM for storing the RTOS image (Image is the common name given for the binary data generated by the compilation of all RTOS source files).
 - ◆ The binary code for RTOS kernel containing all the services is stored in a non-volatile memory (Like FLASH) as either compressed or non-compressed data
 - ◆ During boot-up of the device, the RTOS files are copied from the program storage memory, decompressed if required and then loaded to the RAM for execution
- ◆ FLASH memory is the popular choice for ROM (Program Storage Memory) in embedded applications.
 - ◆ It is a powerful and cost-effective solid-state storage technology for mobile electronics devices and other consumer applications

Memory Selection for Embedded Systems



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Other Embedded Systems component

◆ Sensors

- ◆ A sensor is a transducer device that converts energy from one form to another for any measurement or control purpose.
- ◆ Example for wearable device:
 - ◆ Sensor which counts steps for pedometer functionality is an Accelerometer sensor and
 - ◆ Sensor used in some of the smartwatch devices to measure the light intensity is an Ambient Light Sensor (ALS)

◆ Actuators

- ◆ Actuator is a form of transducer device (mechanical or electrical) which converts signals to corresponding physical action (motion)
- ◆ Actuator acts as an output device
- ◆ Example for wearable device:
 - ◆ smartwatches use Ambient Light Sensor to detect the surrounding light intensity and uses an electrical/ electronic actuator circuit to adjust the screen brightness for better readability

- ◆ For more example you can refer to lab session

Further reading



- ◆ [Embedded Programming](#)
- ◆ [What Is An ASIP?](#)
- ◆ [Specialization Vs. Generalization In Processors](#)
- ◆ [Optimizing efficiency and flexibility in DSP systems](#)
- ◆ [Difference between DSP and General Purpose Processor](#)
- ◆ [Application and type of microprocessor](#)
- ◆ [Testing COTS Systems? Make Evaluation Count](#)
- ◆ [Commercial Off-the-Shelf Components \(COTS\) in Realizing Miniature Implantable Wireless Medical Devices: A Review](#)
- ◆ [What Is an Embedded System and Its Main Components](#)
- ◆ [The Basics of ROM: Types and Uses](#)
- ◆ [Reading and Writing Serial EEPROMs](#)
- ◆ [Erasable Programmable Read Only Memory \(EPROM\)](#)
- ◆ [Real Life Applications of Embedded Systems](#)
- ◆ [How Do Embedded Systems Interact with Various Sensors](#)
- ◆ [Actuators](#)